

MAWSON STATION, Antarctica

WMO: 895640

Lat: 67.602S

Lon: 62.875E

Elev: 16

StdP: 101.13

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-29.0	-27.4	-37.5	0.1	-28.1	-36.0	0.1	-26.8	37.5	-11.6	35.1	-11.3	6.1	160	1.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	3.6	0.2	2.7	-0.5	2.0	-0.9	0.5	3.0	-0.1	2.2	-0.6	1.7	8.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-1.9	3.2	0.9	-2.8	3.0	0.6	-3.5	2.8	0.4	10.6	3.1	9.5	2.3	8.6	1.8	2.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.7	29.1	25.9	DB	-30.3	5.1	2.2	1.1	-31.9	5.9	-33.2	6.6	-34.4	7.2	-36.0	8.0	(4)
(5)			WB	-30.5	1.5	2.1	0.8	-32.1	2.0	-33.4	2.5	-34.6	2.9	-36.1	3.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.3	-0.5	-4.4	-10.4	-14.9	-16.8	-16.4	-17.8	-18.2	-16.9	-13.0	-5.3	-1.0	(6)
(7)		DBStd	7.69	1.65	2.66	3.71	4.51	5.24	5.04	5.72	4.97	4.98	3.91	3.05	1.94	(7)
(8)		HDD10.0	7787	326	404	633	746	831	791	861	876	807	713	460	339	(8)
(9)		HDD18.3	10829	584	637	892	996	1089	1041	1119	1134	1057	971	710	598	(9)
(10)		CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	12.2	9.9	12.3	13.2	12.5	12.7	13.2	13.2	12.9	13.0	11.9	12.0	10.2	(14)
(15)	Precipitation	PrecAvg	311	32	30	25	22	24	26	29	22	21	23	27	29	(15)
(16)		PrecMax	540	80	62	43	56	53	84	69	108	47	67	60	58	(16)
(17)		PrecMin	157	6	8	10	5	6	5	7	4	1	3	2	5	(17)
(18)		PrecStd	86	17	13	10	12	12	16	17	19	13	14	17	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	3.6	-0.5	-3.3	-2.0	-4.3	-4.1	-5.2	-4.5	-2.4	2.0	5.0	(19)
(20)			MCWB	0.9	0.2	-3.0	-5.9	-5.0	-6.7	-5.6	-7.7	-7.0	-5.2	-1.5	1.0	(20)
(21)		2%	DB	3.7	1.5	-2.3	-5.1	-5.4	-6.0	-6.1	-7.8	-6.6	-4.3	0.8	3.7	(21)
(22)			MCWB	0.2	-1.4	-4.6	-7.0	-7.6	-8.0	-7.6	-9.3	-8.5	-6.7	-2.0	0.2	(22)
(23)		5%	DB	2.9	0.5	-3.7	-6.7	-7.5	-7.8	-7.9	-9.2	-8.1	-5.8	0.0	2.8	(23)
(24)			MCWB	-0.3	-2.1	-5.9	-8.3	-9.3	-9.4	-9.4	-10.6	-9.7	-7.9	-2.6	-0.5	(24)
(25)		10%	DB	2.2	-0.3	-5.1	-8.4	-9.3	-9.2	-9.7	-10.9	-9.8	-7.2	-0.9	2.1	(25)
(26)			MCWB	-0.7	-2.8	-7.0	-9.8	-10.6	-10.7	-11.0	-12.2	-11.2	-9.0	-3.3	-0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	0.5	-2.5	-5.4	-4.9	-6.4	-5.2	-7.4	-6.8	-4.9	-0.8	1.4	(27)
(28)			MCDWB	4.1	2.5	-1.0	-3.7	-2.1	-4.5	-4.5	-6.0	-4.9	-2.7	0.9	4.4	(28)
(29)		2%	WB	0.6	-0.8	-4.5	-6.8	-7.4	-7.9	-7.4	-9.3	-8.3	-6.5	-1.6	0.5	(29)
(30)			MCDWB	2.9	1.2	-2.6	-5.1	-5.6	-6.3	-6.2	-7.9	-6.7	-4.5	0.4	3.3	(30)
(31)		5%	WB	0.1	-1.9	-5.7	-8.3	-9.2	-9.3	-9.3	-10.6	-9.7	-7.8	-2.3	-0.2	(31)
(32)			MCDWB	2.4	0.3	-3.9	-6.8	-7.7	-7.8	-8.1	-9.3	-8.2	-5.9	-0.2	2.3	(32)
(33)		10%	WB	-0.4	-2.8	-7.0	-9.8	-10.6	-10.7	-11.0	-12.2	-11.2	-9.0	-3.2	-0.7	(33)
(34)			MCDWB	1.9	-0.6	-5.1	-8.5	-9.2	-9.3	-9.7	-11.0	-9.9	-7.2	-1.0	1.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	4.6	4.3	4.5	4.6	4.5	4.6	4.6	5.1	4.8	4.0	(35)
(36)			MCDBR	4.3	4.5	4.6	4.6	4.8	5.1	4.4	4.8	4.6	5.4	4.7	4.5	(36)
(37)			MCWBR	3.4	3.8	3.7	3.8	4.1	4.4	3.9	4.1	3.9	4.7	4.0	3.6	(37)
(38)		5% WB	MCDWR	3.9	4.1	4.1	4.3	4.7	5.0	4.3	4.7	4.5	5.3	4.4	4.0	(38)
(39)			MCWBR	3.2	3.6	3.5	3.7	4.0	4.3	3.9	4.1	3.9	4.6	3.9	3.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.239	0.212	0.166	0.117	N/A	0.119	0.162	0.198	0.210	0.239	0.258	(40)	
(41)		taud	2.267	2.336	2.326	2.160	1.908	N/A	1.890	2.117	2.229	2.243	2.231	2.214	(41)	
(42)		Ebn at Noon	999	965	895	717	273	0	227	705	893	993	1016	1012	(42)	
(43)		Edh at Noon	114	91	68	39	9	0	8	40	73	99	117	125	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.86	5.44	2.82	0.98	0.12	0.00	0.03	0.53	2.13	4.90	7.63	8.94	(44)	
(45)		RadStd	0.39	0.39	0.21	0.05	0.01	0.00	0.00	0.03	0.09	0.17	0.21	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-0.98	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page